AI Powered Interview Evaluation & Soft Skill Analysis Module

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The rapid growth of online recruitment platforms and Applicant Tracking Systems (ATS) has significantly transformed the hiring process by enabling organizations to manage large volumes of job applications efficiently. However, most traditional recruitment systems rely heavily on keyword-based filtering and rule-based matching methods, which often fail to accurately evaluate the true compatibility between candidates and job roles. These limitations can result in unsuitable candidate recommendations, overlooked qualified applicants, and reduced recruitment efficiency.

This research proposes the development of a Job–Candidate Matching and Explainable AI Engine designed to improve the accuracy and transparency of candidate selection during the recruitment process. The proposed system applies machine learning techniques to analyze candidate resumes and job descriptions in order to identify meaningful similarities between candidate qualifications and job requirements. By extracting relevant features such as skills, experience, education, and job role requirements, the system calculates similarity scores that determine how well a candidate matches a particular job position.

In addition to computing matching scores, the system integrates Explainable Artificial Intelligence (XAI) techniques to provide clear and interpretable explanations for each recommendation. These explanations highlight the key factors that influence the matching results, such as skill alignment, experience relevance, and missing qualifications. This helps recruiters understand the reasoning behind the system's decisions and increases trust in AI-driven recruitment tools.

The proposed methodology involves data preprocessing, feature extraction from resumes and job descriptions, implementation of machine learning–based similarity models, and integration of an explainable AI mechanism to interpret model predictions. The expected outcome of this research is an intelligent recruitment framework that improves candidate ranking accuracy while maintaining transparency in automated decision-making. By combining machine learning and explainable AI techniques, the system aims to support recruiters in making more informed and fair hiring decisions.

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LIST OF ABBREVIATIONS

Table 1 : List of Abbreviations

AI	Artificial Intelligence
NLP	Natural Language Processing
ML	Machine Learning

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1. INTRODUCTION

1.1 Background and Context

The rapid growth of online recruitment platforms and digital job portals has significantly changed the way organizations attract and evaluate candidates. Companies now receive thousands of job applications for a single job position, making the recruitment process more complex and time-consuming. To manage this large volume of applications, many organizations rely on Applicant Tracking Systems (ATS) and automated recruitment tools that help filter and rank candidates based on predefined criteria.

Most traditional recruitment systems use keyword-based matching techniques to compare candidate resumes with job descriptions. These systems search for specific keywords related to required skills, qualifications, and experience. While keyword matching helps reduce the workload of recruiters, it often fails to capture the deeper context of candidate capabilities and job requirements. For example, two candidates may use different terminology to describe the same skill, yet traditional systems may fail to recognize their similarity. As a result, highly qualified candidates may be overlooked, and recruitment decisions may not always reflect the true suitability of applicants.

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have introduced new opportunities to improve recruitment systems. AI-driven models can analyze large amounts of textual data from resumes and job descriptions to identify meaningful relationships between candidate skills and job requirements. These technologies enable more intelligent job–candidate matching by considering semantic similarity and contextual information rather than relying solely on keyword matching.

However, many AI-based recruitment models still suffer from a lack of transparency. Recruiters often cannot clearly understand how the system determines whether a candidate is suitable for a specific job role. This lack of interpretability can reduce trust in automated recruitment systems and may create concerns about fairness and bias. Therefore, integrating Explainable Artificial Intelligence (XAI) into recruitment systems is important for providing clear explanations about why certain candidates are recommended.

communication style or reactions to situations. Because of these hiring decisions often depend on personal judgment which may lead to bias.

New technologies, like AI and NLP can help solve some problems like these. These technologies can analyze things such as what people say in interviews, how they say it and how

they feel. Also, they can give companies an idea of who someone is and how they might fit in at the company. By using these technologies companies can make the hiring process fairer and more based on facts.

To deal with these issues this project is proposing a system that uses artificial intelligence to help evaluate job interviews and figure out if someone has the right soft skills. The system can look at what people say in interviews. How are there soft skills. This will help companies make better hiring decisions that are based on more than a resume. This project helps promote fair and inclusive hiring. It supports equal job opportunities by using AI to make the recruitment process more transparent and reduce bias. It ensures that every candidate has a fair chance to get a job.

From a research standpoint this project fits into the area of intelligence and data science especially when it comes to using machine learning and data to help companies make better decisions. The project adds to this area of research by looking at how artificial intelligence can be used to analyze job interviews and figure out if someone has soft skills. Recruitment is a part of this, and the project wants to make it better. The goal is to make the hiring process fairer and more effective and to help companies find people for the job.

1.2 Literature Review

The rapid advancement of digital technologies has significantly transformed the recruitment and hiring landscape. Traditionally, recruitment relied heavily on manual resume screening, recruiter intuition, and face to face interviews to assess candidate suitability. However, the emergence of Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) has enabled organizations to automate and enhance various aspects of the recruitment process. Modern AI driven recruitment systems aim to improve efficiency, reduce hiring time, and provide more objective evaluation of candidate capabilities.

Recent research highlights the increasing adoption of AI powered interview systems that utilize advanced computational techniques to analyze candidate responses, communication patterns, and behavioral characteristics. These systems attempt to simulate human like interview experiences by combining multiple technologies such as speech recognition, sentiment analysis, and computer vision. For example, Tighare et al. [1] introduced the Skill Sight interview assessment platform, which focuses on evaluating candidates' emotional intelligence by analyzing facial expressions, reaction time, and behavioral responses during interviews. The study emphasizes that nonverbal communication cues such as facial expressions and eye movements provide valuable indicators of a candidate's emotional awareness and confidence during professional interactions.

Similarly, Jagtap et al. [4] and Patil et al. [8] proposed AI driven interview simulation platforms that utilize real time voice based agents to conduct automated interview sessions. These systems employ Speech to Text (STT) technologies combined with conversational AI models to interpret candidate responses and generate follow up questions dynamically. Such systems attempt to replicate natural conversation patterns, thereby creating a more interactive and realistic interview experience. These frameworks also enable automated response evaluation by analyzing linguistic features such as vocabulary usage, sentence complexity, and contextual relevance.

Several studies have also explored the integration of deep learning techniques for candidate behavior analysis. Emotion recognition models based on Convolutional Neural Networks (CNNs) have been widely used to analyze facial expressions and emotional states during video interviews. These models can identify emotions such as confidence, nervousness, or uncertainty by analyzing facial landmarks and micro expressions [5]. In addition, advanced machine learning models such as XGBoost classifiers and Transformer based architectures have been applied to evaluate candidate responses and generate structured feedback for recruiters [10]. These approaches enable the extraction of meaningful patterns from candidate responses and allow systems to evaluate interview performance more systematically.

Furthermore, some recent research has explored the concept of immersive AI interview environments using advanced human computer interaction technologies. Wu et al. [7] proposed an AI interview evaluation framework that incorporates Virtual Digital Humans (VDH) to simulate human interviewers. These digital avatars interact with candidates in real time and provide a more engaging and realistic interview environment. The use of multi agent systems within such platforms allows dynamic question generation, behavioral analysis, and adaptive feedback mechanisms. These approaches demonstrate the potential of integrating AI driven conversational agents with advanced visualization technologies to create more interactive recruitment systems.

In addition to academic research, several commercial platforms have emerged that aim to modernize recruitment processes. Platforms such as HireVue, Spark Hire, and VidCruiter provide video interviewing solutions that allow recruiters to conduct remote interviews and record candidate responses for later evaluation [1]. These platforms integrate basic Applicant Tracking System (ATS) functionality, enabling organizations to store candidate data, manage interview scheduling, and track recruitment workflows. Similarly, specialized platforms such as Codility focus on technical skill assessment by providing automated coding tests and performance evaluation tools. Another example is Google's Interview Warmup, which is designed to help candidates practice interview questions and receive automated feedback on their responses.

Despite these technological advancements, current recruitment systems still exhibit several limitations. Many existing platforms primarily focus on technical skill assessment or video recording capabilities, without performing deep behavioral analysis of candidate responses. According to systematic reviews conducted by Sanjana et al. [2], [6], most AI driven recruitment systems lack integrated frameworks that simultaneously analyze linguistic content, emotional cues, and behavioral characteristics during interviews. As a result, these systems often provide limited insights into candidate communication abilities and interpersonal skills, which are critical for many professional roles.

Another key limitation identified in existing research is the lack of explainability and transparency in AI based candidate evaluation. In many automated recruitment systems, AI models generate candidate scores without providing clear explanations of how these scores were calculated. This lack of interpretability may lead to concerns regarding fairness, bias, and accountability in recruitment decisions [7], [10]. Furthermore, many platforms rely heavily on keyword based resume filtering techniques, which may overlook contextual skills and transferable experience [3], [9]. Such approaches may result in the rejection of highly qualified candidates who do not use specific keywords in their resumes.

Additionally, several studies highlight the absence of personalized feedback mechanisms within existing AI interview systems. While many platforms provide automated interview scoring, they rarely

offer detailed insights or learning recommendations that help candidates improve their performance [2], [6]. The absence of such pedagogical support limits the effectiveness of these systems as career development tools.

Overall, although current research demonstrates significant progress in AI powered recruitment technologies, existing systems still lack a holistic evaluation framework that integrates communication analysis, emotional intelligence detection, and explainable AI based scoring within a single platform. This limitation highlights the need for more comprehensive recruitment solutions that evaluate both technical competency and soft skills in an objective and transparent manner.

1.2 Research Gap

Although many recruitment systems and research studies have explored automated candidate screening and job–candidate matching, several limitations still exist in current approaches. Traditional Applicant Tracking Systems (ATS) mainly rely on keyword-based filtering techniques to compare candidate resumes with job descriptions. These systems often fail to capture the deeper semantic meaning of candidate skills, experience, and qualifications. As a result, candidates who possess relevant abilities but use different terminology in their resumes may be overlooked, leading to inaccurate candidate shortlisting.

Recent studies have attempted to improve job–candidate matching using machine learning and Natural Language Processing (NLP) techniques. These approaches analyze textual information from resumes and job descriptions to identify relationships between candidate qualifications and job requirements. While such methods improve the accuracy of candidate matching compared to simple keyword filtering, many existing systems still focus primarily on similarity scoring without providing sufficient explanations for their recommendations.

Another major limitation of many AI-based recruitment systems is the lack of transparency in their decision-making process. Most machine learning models operate as black-box systems, where recruiters cannot clearly understand why a candidate has been recommended or rejected. This lack of interpretability reduces trust in automated recruitment systems and makes it difficult for recruiters to justify hiring decisions based on AI-generated results.

Furthermore, many current research solutions address job matching and candidate ranking independently, without integrating explainable mechanisms that clearly highlight the factors influencing the matching results. Recruiters often require detailed insights such as which skills contributed to a high match score or which requirements are missing from a candidate profile.

Therefore, there is a need for an intelligent recruitment system that not only improves the accuracy of job–candidate matching using machine learning techniques but also provides transparent and interpretable explanations for the recommendations. Addressing this gap will help recruiters better understand AI-driven decisions, increase trust in automated recruitment systems, and support more effective and fair hiring processes.

OBJECTIVE

2.1 Main Objective

The main objective of this research is to design and develop a **machine learning–based Job–Candidate Matching and Explainable AI Engine** that can accurately analyze candidate resumes and job descriptions in order to determine the suitability of candidates for specific job roles. The proposed system aims to improve the recruitment process by calculating similarity scores between candidate profiles and job requirements using intelligent matching algorithms.

In addition to improving the accuracy of candidate matching, the system will integrate **Explainable Artificial Intelligence (XAI)** techniques to provide clear and interpretable explanations for the recommendations generated by the model. These explanations will highlight the key factors influencing the matching results, such as relevant skills, experience alignment, and missing job requirements.

The ultimate goal of this research is to support recruiters in making more informed and transparent hiring decisions by combining machine learning–based candidate matching with explainable insights that increase trust and reliability in automated recruitment systems.

2.2 Specific Objectives

• Data Collection and Preprocessing
• Collect candidate resumes and job descriptions from available recruitment datasets or online job portals.
• Perform data preprocessing by cleaning textual data, removing noise, and preparing the dataset for machine learning analysis.

• Resume and Job Description Feature Extraction
• Extract important features such as candidate skills, experience, education, and job requirements using Natural Language Processing (NLP) techniques.
• Convert textual data into structured representations suitable for similarity analysis.

• Job–Candidate Matching Model Development
• Implement machine learning or similarity-based algorithms to calculate compatibility between candidate profiles and job descriptions.
• Generate matching scores that represent how closely a candidate's qualifications align with the job requirements.

• Explainable AI Integration
• Apply Explainable Artificial Intelligence (XAI) techniques to interpret the model's predictions.
• Provide human-readable explanations highlighting key factors influencing candidate recommendations.

• System Evaluation and Performance Analysis
• Evaluate the performance of the matching model using metrics such as accuracy, precision, recall, and ranking effectiveness.
• Analyze how effectively the system recommends suitable candidates and improves recruitment decision-making.

3. METHODOLOGY

This research focuses on developing a machine learning–based Job–Candidate Matching and Explainable AI Engine that improves the accuracy and transparency of recruitment systems. The proposed methodology follows a structured process consisting of data collection, preprocessing, feature extraction, model development, and explanation generation to support intelligent candidate matching.In the first stage, recruitment-related data such as candidate resumes and job descriptions are collected from available datasets or online recruitment platforms. These datasets contain important information including candidate skills, education, work experience, and job requirements. The collected data is stored and organized to support further analysis.

The second stage involves data preprocessing, where raw textual data from resumes and job descriptions is cleaned and prepared for analysis. This includes removing irrelevant information, handling missing data, normalizing text, and converting unstructured textual information into a consistent format. Natural Language Processing (NLP) techniques are applied to extract meaningful features such as candidate skills, experience duration, education level, and job requirement keywords.After preprocessing, the extracted features are used to represent candidate profiles and job descriptions in a structured format suitable for machine learning models. Techniques such as vectorization or embedding-based representations are applied to convert textual information into numerical form. These representations allow the system to measure semantic similarity between candidate qualifications and job requirements.

In the next stage, machine learning–based similarity models are implemented to compute job–candidate compatibility scores. Algorithms such as cosine similarity, machine learning classification models, or ranking algorithms may be used to analyze relationships between candidate features and job requirements. The system then ranks candidates based on their calculated compatibility scores.

To improve transparency and interpretability, Explainable Artificial Intelligence (XAI) techniques are integrated into the system. These techniques analyze the contribution of different features in the prediction process and generate explanations that highlight the factors influencing candidate recommendations. For example, the system may explain that a candidate received a high match score due to relevant technical skills, years of experience, or educational background.

Finally, the generated matching scores and explanations are presented through a decision-support interface that allows recruiters to review candidate rankings and understand the reasoning behind each recommendation. This methodology ensures that the proposed

system not only improves the accuracy of job–candidate matching but also enhances transparency and trust in AI-driven recruitment decisions.

3.3 Tools and Platforms

The system will be implemented using a combination of modern programming tools, machine learning frameworks, and cloud technologies to ensure scalability and reliability.
The primary development environment will use Python, which provides extensive libraries and frameworks for artificial intelligence and natural language processing. The OpenAI Whisper model will be used for speech recognition tasks, while SpaCy and Hugging Face Transformers will support advanced language analysis.

The backend services will be implemented using Flask or FastAPI, enabling the system to expose APIs for interview processing and data management. A web based dashboard interface, potentially developed using React or similar web technologies, will provide recruiters with visualization tools to review candidate evaluations and reports.

For scalable processing and deployment, the system may utilize cloud computing platforms such as Amazon Web Services (AWS) or Google Cloud Platform (GCP). These platforms provide computational resources required for machine learning processing and enable the system to support multiple users simultaneously.

The selected technologies are widely used in AI based applications due to their reliability, scalability, and strong ecosystem support for machine learning and NLP development.

3.4 Data Collection and Ethical Considerations

The data required for this research will consist of candidate resumes, job descriptions, and recruitment datasets. These datasets will be obtained from publicly available sources and synthetic data generated to simulate realistic hiring scenarios.

To ensure data privacy and responsible usage, all datasets will be anonymized before processing. Any personally identifiable information such as names, contact details, and sensitive attributes will be removed to maintain confidentiality.

Since the proposed system processes candidate profiles and recruitment-related data, ethical considerations are an important aspect of this research. Ethical practices such as secure data storage, controlled access to recruitment information, and responsible use of AI models will be strictly followed. Additionally, all data will be used solely for research purposes, and compliance with ethical data handling guidelines will be ensured throughout the study.

3.5 Validation Strategy and Performance Metrics

The performance and reliability of the proposed job–candidate matching system will be evaluated using multiple metrics to measure the accuracy and effectiveness of the matching and ranking process.

The semantic matching component will be evaluated by measuring the accuracy of similarity scores between candidate profiles and job descriptions. This ensures that the system correctly identifies relevant candidates based on their skills and experience.

The candidate ranking module will be assessed using evaluation metrics such as Precision, Recall, and F1-score to determine how effectively the system ranks suitable candidates for a given job role.

The Explainable AI component will be evaluated based on the clarity and usefulness of the generated explanations, ensuring that key factors influencing candidate recommendations are understandable to recruiters.

Finally, the overall system performance will be validated by comparing AI-generated candidate rankings with manual evaluations performed by HR professionals. Correlation analysis will be used to measure the agreement between automated recommendations and human judgments, indicating the reliability and effectiveness of the system.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and Its Components

The proposed system is a **Job–Candidate Matching & Explainable AI Engine**, which functions as a core component of the **AI-Powered Intelligent Talent Acquisition System**. This module is designed to automatically match candidate profiles with job descriptions and generate explainable recommendations to support objective recruitment decisions.

Traditional recruitment systems mainly rely on keyword-based matching techniques, which often fail to capture the semantic relationships between candidate skills and job requirements. This leads to inaccurate candidate selection and a lack of transparency in decision-making. The proposed system addresses these limitations by integrating **Natural Language Processing (NLP), Machine Learning, and Explainable AI (XAI)** to provide accurate and interpretable job–candidate matching.

The architecture consists of several interconnected components that work together to process candidate and job data and generate matching results. These components include:

Candidate & Job Data Input Module

This component collects structured candidate profiles (generated from resume parsing) and job descriptions. It acts as the input layer of the system.

Text Preprocessing Module

Candidate profiles and job descriptions are processed using NLP techniques such as tokenization, normalization, and stop-word removal to prepare the data for analysis.

Semantic Embedding Generation Module

This module converts textual data into numerical vector representations using embedding models such as Sentence-BERT. These embeddings capture the semantic meaning of candidate qualifications and job requirements.

Semantic Job–Candidate Matching Engine

This component computes similarity scores between candidate profiles and job descriptions using machine learning–based similarity models. It evaluates skill relevance and experience alignment to determine candidate suitability.

Explainable AI (XAI) Analysis Module

This module provides interpretable insights into the matching results by identifying key factors influencing candidate recommendations, such as matched skills and missing requirements.

Candidate Ranking & Recommendation Module

Candidates are ranked based on their similarity scores, and recommendation insights are generated to assist recruiters in selecting the most suitable candidates.

Recruitment Decision Support Dashboard

The final outputs, including ranked candidate lists and explainable insights, are presented through a dashboard interface to support data-driven hiring decisions.

By integrating these components, the system provides an **intelligent, transparent, and efficient job–candidate matching framework** that improves recruitment accuracy, reduces bias, and enhances decision-making in hiring processes.

4.2 High-Level System Architecture Diagram

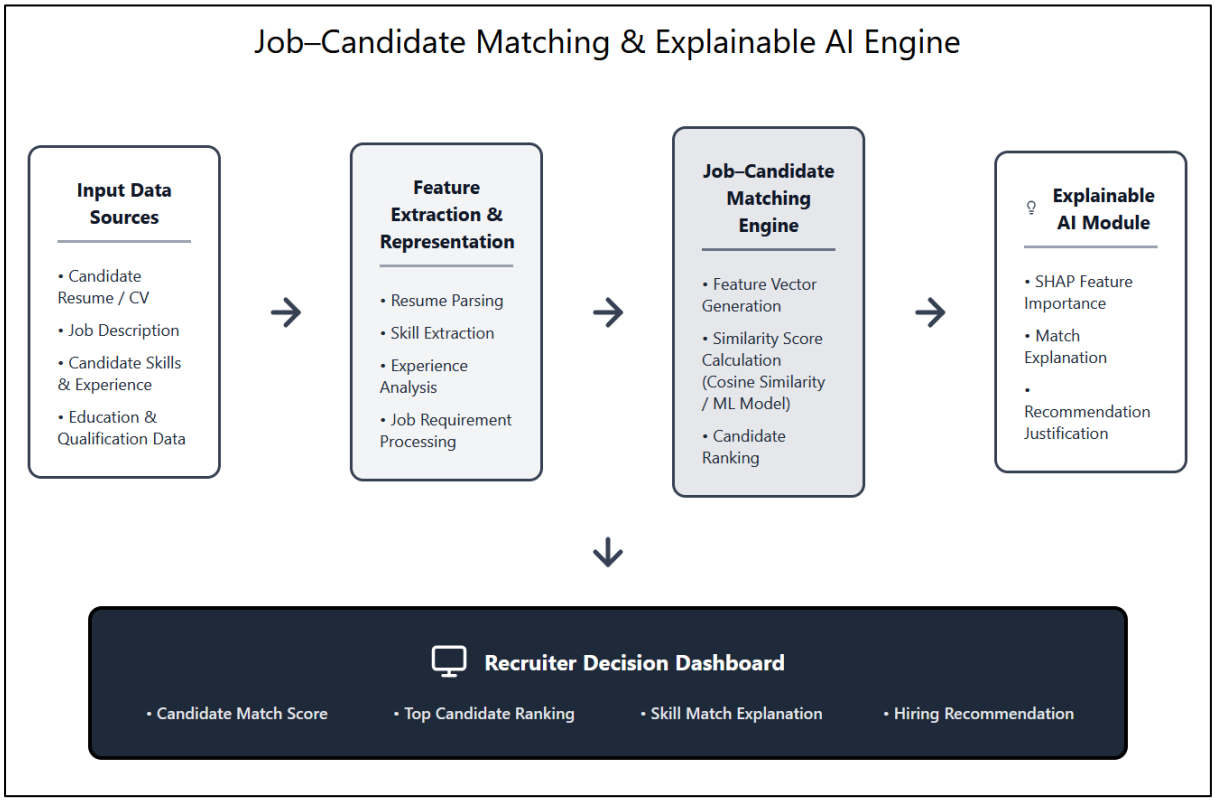


Fig.1. ***Job–Candidate Matching & Explainable AI Engine***

Figure 1 illustrates **the high-level architecture of the Job–Candidate Matching & Explainable AI Engine within the AI-Driven Intelligent Recruitment Decision Platform.** The system operates as part of the AI Analysis Layer and is responsible for accurately matching candidate profiles with job requirements and providing explainable insights.

The process begins after candidate data is collected and processed into structured profiles within the Candidate Data Repository. The Job–Candidate Matching Engine then retrieves candidate profiles and job descriptions to perform semantic analysis.

The system first applies **NLP-based preprocessing and embedding techniques** to convert textual information into numerical representations. Using these representations, the matching engine computes **semantic similarity scores** between candidate qualifications and job requirements, evaluating factors such as skill relevance and experience alignment.

To enhance transparency, the system integrates an **Explainable AI (XAI) module**, which identifies the key factors influencing matching results. This includes highlighting matched skills, missing requirements, and feature contributions that affect candidate ranking.

The final outputs are passed to the **Recruitment Decision Dashboard**, where recruiters can view ranked candidate lists, similarity scores, and explainable insights to support informed hiring decisions.

Overall, this component enables **accurate, transparent, and data-driven job–candidate matching**, improving recruitment efficiency and reducing bias in the hiring process.

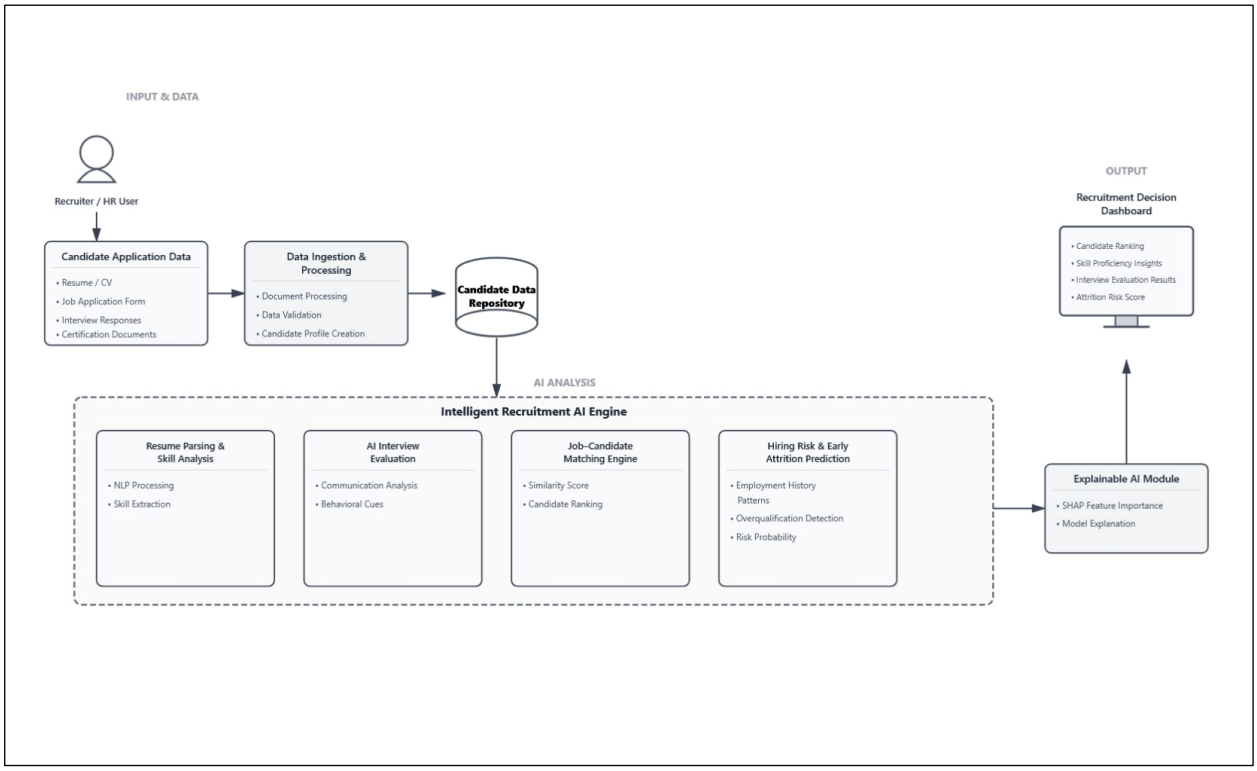


Fig.2. ***Job–Candidate Matching & Explainable AI Engine***

Figure 2 illustrates **the detailed architecture of the Job–Candidate Matching & Explainable AI Engine.** The system is designed to process candidate profiles and job descriptions through multiple interconnected components, including text preprocessing, semantic embedding generation, similarity computation, and explainable AI analysis.

Each module performs a specific function, contributing to accurate and intelligent job–candidate matching. The semantic matching engine calculates similarity scores by analyzing skill relevance and experience alignment between candidate profiles and job requirements.

The Explainable AI module interprets the matching results by identifying key factors such as matched skills and missing requirements. These outputs are then used by the ranking module to generate a prioritized list of candidates.

4.3 Component Interactions

- The system components interact sequentially to process candidate and job data and generate matching insights. When candidate profiles and job descriptions are provided, the data is first passed to the text preprocessing module, where NLP techniques are applied to clean and normalize the text.

- The processed data is then forwarded to the semantic embedding module, which converts candidate profiles and job descriptions into numerical vector representations. These embeddings are used by the job–candidate matching engine to compute similarity scores based on skill relevance and experience alignment.

- The Explainable AI module then analyzes the matching results to identify key factors influencing the recommendations, such as matched skills and missing requirements. These insights help improve transparency in the decision-making process.

- The ranking module aggregates the similarity scores and generates a prioritized list of candidates based on their suitability for the job role.

- Finally, the results are presented through the recruiter dashboard, where recruiters can view ranked candidates, matching scores, and explainable insights to support data-driven hiring decisions.

4.4 Data Flow Explanation

- The system follows a structured data processing pipeline.

1.Data Input Stage

Candidate profiles and job descriptions are collected through the recruitment platform.

2.Text Preprocessing Stage

The input data is processed using NLP techniques to clean and normalize the text.

3.Semantic Representation Stage

Candidate profiles and job descriptions are converted into vector representations using embedding models.

4.MatchingStage

The job–candidate matching engine computes similarity scores based on skill relevance and experience alignment**.**

5.Explainable AI Stage

The system analyzes key factors influencing the matching results, such as matched skills and missing requirements.

6.Ranking Stage

Candidates are ranked based on their similarity scores and suitability for the job role.

7.OutputStage

Recruiters receive ranked candidate lists, matching scores, and explainable insights through the dashboard.

• This data flow enables the system to transform candidate and job data into accurate and transparent recruitment insights.

4.5 Scalability and Security Considerations

- To support large-scale recruitment environments, the proposed job–candidate matching system incorporates several scalability and security features.

Scalability

- The system is designed using a modular architecture, allowing the job–candidate matching engine and explainable AI module to operate independently. Cloud platforms such as AWS or Google Cloud can be used to deploy scalable services that handle multiple candidate-job matching processes simultaneously.

- Containerization technologies such as Docker can be used to ensure flexible deployment and efficient resource utilization. This enables the system to scale based on recruitment demand and data volume.

Security

- Since the system processes sensitive candidate and recruitment data, strong security mechanisms are essential.

- All candidate profiles and job-related data will be encrypted during transmission and storage to ensure data protection. Access control mechanisms will restrict system usage to authorized recruiters and HR personnel.

- Additionally, secure authentication methods and role-based access control will be implemented to prevent unauthorized access. Data anonymization techniques will also be applied to protect candidate privacy and ensure compliance with ethical data handling standards.

5. USER REQUIREMENTS

5.1 Stakeholders

- **Recruiters and HR Professionals**
  - Primary users who manage the hiring process.
  - They rely on the system to screen candidates objectively, evaluate soft skills, and access detailed interview reports.

- **Hiring Managers**
  - Oversee department level recruitment decisions.
  - They use the system's candidate readiness scores and behavioral insights to make informed hiring choices.

- **Job Candidates**
  - Individuals participating in interviews.
  - They interact with the system to record or submit responses, and benefit from fair, unbiased evaluation and possible feedback.

- **System Administrators**
  - Responsible for maintaining the system, managing user accounts, ensuring data security, and supporting cloud or server infrastructure.

5.2 Functional Requirements

FR1 - Candidate & Job Data Input

The system shall accept candidate profiles and job descriptions for processing.

FR2 - Text Preprocessing

The system shall preprocess textual data using NLP techniques for analysis.

FR3 - Semantic Representation

The system shall convert candidate profiles and job descriptions into vector representations using embedding models.

FR4 - Job–Candidate Matching

The system shall compute similarity scores between candidates and job requirements using machine learning models.

FR5 - Skill & Experience Alignment

The system shall evaluate skill relevance and experience alignment to determine candidate suitability.

FR6 - Explainable AI Insights

The system shall generate interpretable explanations highlighting matched skills and missing requirements.

FR7 - Candidate Ranking

The system shall rank candidates based on similarity scores and suitability.

- **FR8 - Reporting and Dashboard**

  Provide recruiters with detailed visual reports, including explanations of scores and behavioral insights.

5.3 Non-Functional Requirements

Security - All candidate profiles and job-related data must be encrypted and securely stored to prevent unauthorized access and ensure data privacy.

Performance - The system should efficiently process candidate-job matching and generate similarity scores and recommendations within a short time.

Scalability - The system must handle large volumes of candidate profiles and job descriptions simultaneously without performance degradation.

Usability - Recruiters should be able to easily access candidate rankings, matching insights, and explanations through a user-friendly dashboard.

Reliability - The system must provide consistent and accurate matching results with high availability to ensure dependable recruitment decisions.

6. COMMERCIALIZATION PLAN

The proposed **Job–Candidate Matching & Explainable AI Engine** targets a wide range of users, including small and medium enterprises, corporate HR departments, recruitment agencies, and online hiring platforms. These organizations often struggle to efficiently match candidates with job roles while ensuring accuracy and fairness, making this system highly valuable.

The value proposition of the system lies in its ability to provide **accurate, data-driven, and transparent job–candidate matching**. By using semantic similarity and Explainable AI, the system improves candidate selection, reduces recruiter workload, and enhances trust in AI-driven hiring decisions.

The revenue model is designed to be flexible and scalable. Organizations can adopt a **Software-as-a-Service (SaaS) subscription model**, purchase enterprise licenses, or use a pay-per-use model based on the number of candidate-job matches processed. Costs include system development, cloud infrastructure, and ongoing maintenance. A tiered pricing strategy ensures accessibility for both small businesses and large enterprises.

The system offers several competitive advantages over traditional recruitment tools. These include **semantic job–candidate matching beyond keyword filtering, context-aware skill alignment, explainable AI insights for transparency, and intelligent candidate ranking mechanisms**. These features differentiate the system from conventional Applicant Tracking Systems.

Finally, in terms of intellectual property, the matching algorithms, explainable AI framework, and system architecture can be protected through software copyright and potential patent registration, ensuring innovation protection and commercial value.

7. BUDGET AND JUSTIFICATION

Table 2: Estimated expenses of the proposed system

Category	Requirement/ Resource	Description	Estimated Cost (LKR)
Hardware	High-Performance Development Laptop	Used for implementing AI models, running NLP algorithms, and processing interview audio data	230,000
Hardware	External Backup Storage (SSD)	Used for storing interview recordings, processed datasets, and trained model files	18,000
Hardware	Audio Recording Equipment	Microphone and headset used for collecting sample interview datasets for testing	12,000
Software	Programming Environment (Python, VS Code)	Development environment used for building backend services and AI modules	0 (Open Source)
Software	AI and NLP Frameworks	Libraries such as Whisper, SpaCy, HuggingFace Transformers, and TextBlob used for AI analysis	0 (Open Source)
Software	Data Processing Libraries	Tools such as Pandas, NumPy, and Scikit-learn used for data preprocessing and evaluation	0 (Open Source)
Cloud Services	Cloud Computing Instance	Cloud-based GPU/CPU resources used for processing interview data and running AI models	35,000
Cloud Services	Cloud Storage and Database	Secure storage for interview recordings, transcripts, candidate profiles, and system logs	22,000
Human Effort	System Design and Development	Research, implementation, testing, and documentation conducted as part of academic project work	0
Total Estimated Cost			**317,000 LKR**

The estimated budget for the proposed system primarily focuses on computational infrastructure and data processing resources required to support artificial intelligence development. The high-performance development laptop represents the primary hardware requirement for the project. Machine learning models used for speech recognition and natural

language processing require significant computational capacity during development and testing. The laptop will support tasks such as training models, processing interview recordings, and running AI evaluation algorithms. An external solid-state storage device is included to ensure secure storage of datasets, system backups, and trained model files. Since interview recordings and transcription datasets can consume large amounts of storage space, external backup storage is necessary for safe data management. The system also requires audio recording equipment to collect sample interview responses for training and testing purposes. These recordings help simulate real interview scenarios and allow evaluation of the speech-to-text and communication analysis components. Most software tools used in the project are open-source technologies, including Python programming language, development environments such as Visual Studio Code, and AI frameworks such as Whisper, SpaCy, HuggingFace Transformers, and TextBlob. These tools significantly reduce development costs while providing advanced capabilities for speech recognition, natural language processing, and sentiment analysis. The proposed system will also utilize cloud computing services for scalable deployment and testing. Cloud infrastructure enables the system to process interview recordings, run AI inference tasks, and support multiple users simultaneously. Cloud storage services will securely store candidate data, interview transcripts, and analysis results while ensuring data availability and reliability. Finally, the human effort cost associated with system design, development, testing, and research activities is not included in the financial estimate because the work is carried out as part of the academic research project. Overall, the estimated budget of 317,000 LKR is sufficient to support the development and evaluation of a functional prototype of the AI-based interview evaluation system while maintaining cost efficiency with open-source technologies.

8. WORK BREAKDOWN STRUCTURE (WBS)

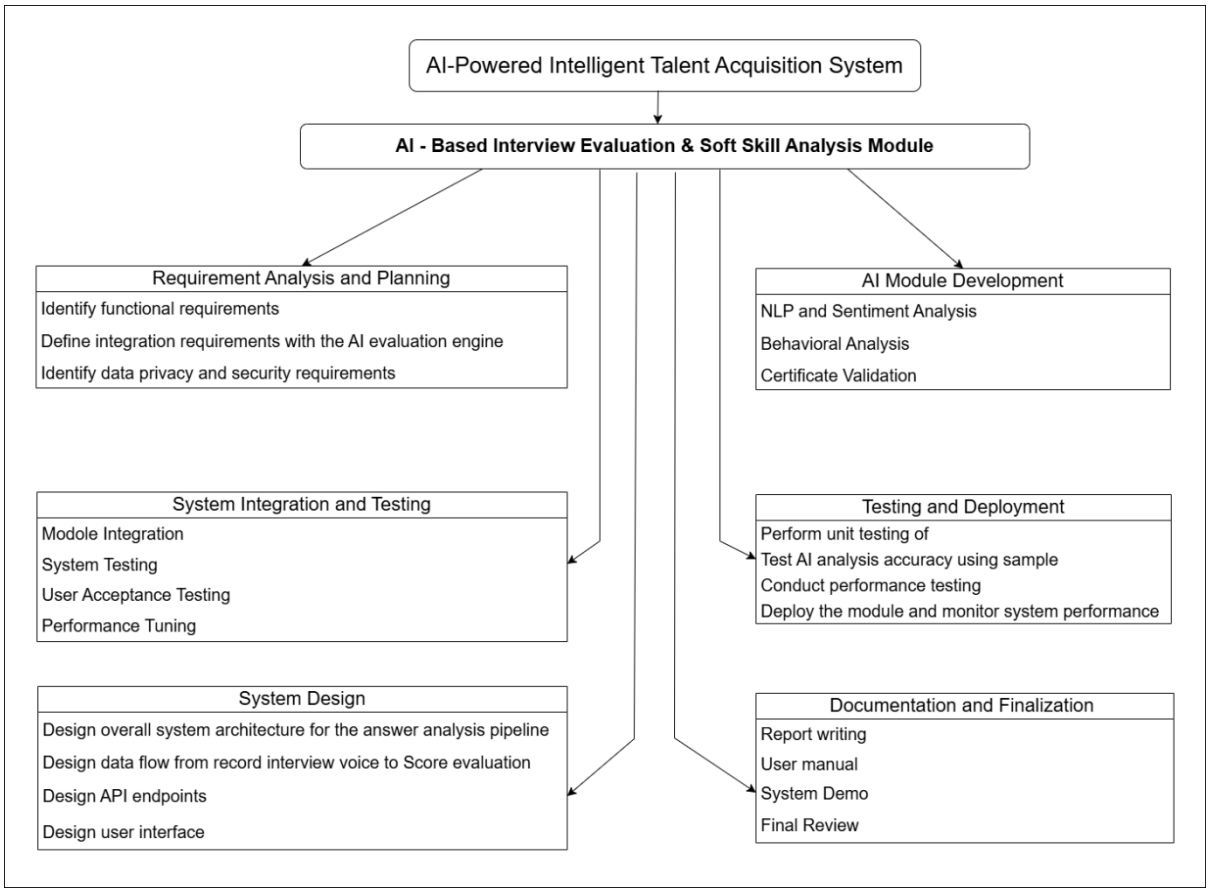


Fig.3. Work Breakdown Structure

Figure 3 Work Breakdown Structure (WBS) diagram provides a hierarchical representation of the project tasks. It visually organizes the research and development activities into manageable components, showing dependencies and timelines for each task. The WBS ensures balanced workload distribution, facilitates tracking of project progress, and helps align individual responsibilities with overall project objectives for the AI-Based Interview Evaluation and Soft Skill Analysis system.

9. GANTT CHART

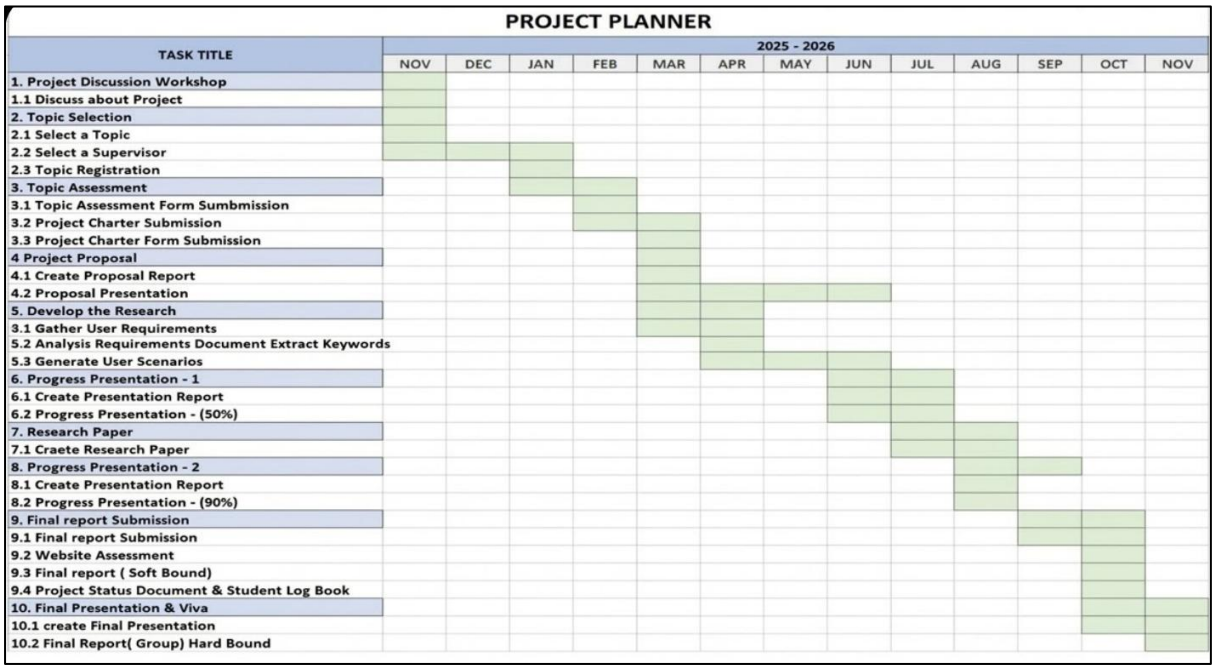


Fig.4. Gantt Chart

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11.APPENDICES

Appendices 1:Risk analysis

Risk	Description	Impact	Likelihood
Inaccurate Matching Results	Semantic matching may not fully capture candidate suitability due to incomplete or ambiguous data.	Medium	Medium
NLP Misinterpretation	NLP models may incorrectly interpret skills or experience from candidate profiles.	Medium	Medium
Data Quality Issues	Incomplete or poorly structured resumes and job descriptions may affect matching accuracy.	High	Medium
Data Privacy Breaches	Candidate profiles and recruitment data may be exposed if security measures are insufficient.	High	Low
System Performance Issues	Processing large volumes of candidate-job matching may increase response time.	Medium	Medium
Model Overfitting	Matching models trained on limited datasets may not generalize well to real-world scenarios.	Medium	Medium
Integration Failures	Issues in integration with other modules such as resume parsing or dashboard may affect system functionality.	Medium	Low
Bias in Matching Decisions	AI models may introduce bias if training data is not balanced or representative.	High	Low